

Project Design Phase
Solution Architecture

Date	10 February 2026
Team ID	LTVIP2026TMIDS90611
Project Name	Travel-Itinerary-Generator
Maximum Marks	4 Marks

Solution Architecture:

1. User Interface Layer (Streamlit Web UI)

- This layer provides the interface through which users interact with the application.
 - Travelers can enter travel preferences such as destination, travel dates, budget, and interests.
 - The interface displays the generated itinerary and allows users to modify or download it.
 - Built using **Streamlit**, which enables interactive web applications with minimal code.
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2. Application Logic Layer (Python Backend)

- The backend processes user inputs received from the web interface.
 - It manages the application logic such as validating user inputs and preparing requests for itinerary generation.
 - Handles communication between the user interface, AI processing layer, and database.
 - Implemented using **Python** for efficient data processing.
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3. AI Processing Layer (Gemini API)

- This layer uses **Gemini AI API** to generate intelligent travel recommendations.
 - The AI analyzes user preferences and suggests destinations, attractions, and travel schedules.
 - It creates a structured and personalized travel itinerary.
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4. Data Management Layer

- Stores important data such as user accounts, destinations, and generated itineraries.
- Enables users to save, retrieve, and update their travel plans.
- Ensures efficient data retrieval and management.

5. Output Rendering Layer

- Displays the generated itinerary in a structured and user-friendly format.
- Provides options to view, modify, or download the itinerary.
- Presents recommendations clearly for better travel planning.

Technology Stack

- **Streamlit Web UI** – Frontend interface
- **Python Backend** – Application logic and processing
- **Gemini AI API** – Intelligent itinerary generation
- **Deployment** – Cloud or local hosting for application availability

Example - Solution Architecture Diagram:

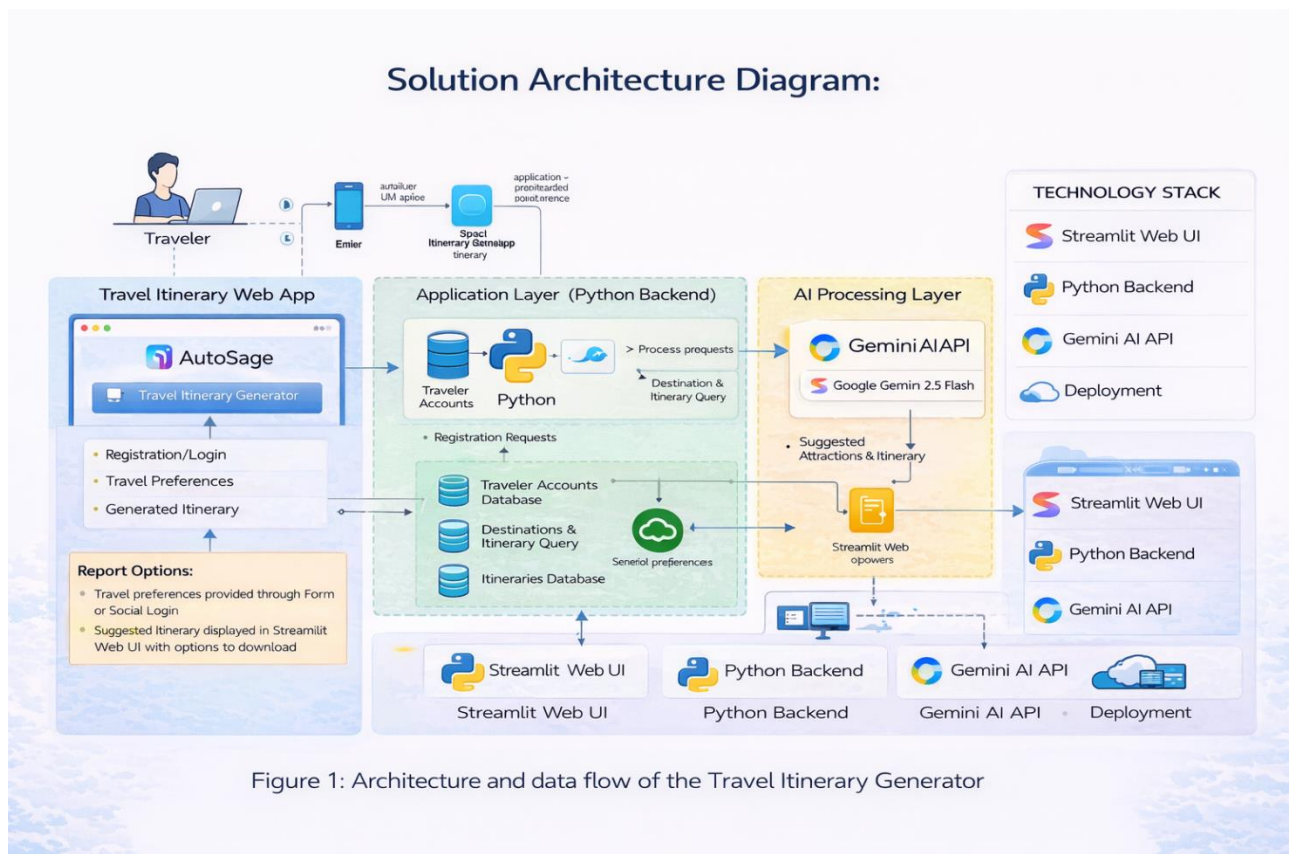


Figure 1: Architecture and data flow of the Travel Itinerary Generator

Figure 1: Architecture and data flow of the voice patient diary sample application